A

Research Paper

On

Walmart Sales Data Analysis and Forecasting Dashboard

in partial fulfilment of the requirement for the degree of

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

Submitted By :-

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STUDENT'S DECLARATION

I hereby certify that the work which is being presented in the major project report entitled "Walmart Sales Data Analysis and Forecasting Dashboard" in fulfillment of the requirement for the award of the Degree of Bachelor of Technology in Department of Data Science of Noida Institute of Engineering and Technology, Greater Noida ,Uttar Pradesh is an authentic record of my own work carried out during 8th semester.

Date: 17/3/2026

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The major project viva-voce examination of Ms. Barkha Jha, Roll No 2201331540064 Of B.TECH (Branch) has been held on _____.

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Noida Institute of Engineering & Technology**, Department of Computer Science and Engineering (Data Science), for providing me with the opportunity to work on the project titled **"Walmart Sales Data Analysis and Forecasting Dashboard"**.

I am deeply thankful to my project guide **Ms. Garima Dhawan,** for his valuable guidance, continuous encouragement, and constructive suggestions throughout the development of this project. His support helped me successfully complete this work.

I would also like to extend my gratitude to all faculty members of the Department of CSE (Data Science) for their continuous support and motivation during the course of this project.

I would also like to thank my friends and classmates for their suggestions, cooperation, and encouragement during the project development.

Finally, I express my heartfelt gratitude to my family for their constant support, motivation, and encouragement throughout this project.

This project has helped me gain practical knowledge in **Data Analysis, Machine Learning, Power BI, and Data Visualization**, and enhanced my technical and analytical skills

Barkha Jha

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CERTIFICATE

This is to certify that the research work entitled
"Walmart Sales Data Analysis and Forecasting using Machine Learning and Business Intelligence"
is a bonafide work carried out by **Barkha Jha (Roll No: 2201331540064)** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering (Data Science)** from **Noida Institute of Engineering & Technology, Greater Noida**. This work has been carried out under my supervision and guidance during the academic session. The work presented in this report has not been submitted elsewhere for the award of any other degree or diploma.
I hereby certify that the above statement is true to the best of my knowledge.

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Walmart Sales Data Analysis And Forecasting Using Machine Learning And Business Intelligence

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ABSTRACT

Retail businesses generate large volumes of transactional data from daily operations. Analyzing this data is essential for understanding customer behavior, product demand, and overall business performance. However, extracting meaningful insights from raw sales data remains a major challenge for retail organizations. This research paper presents a **Walmart Sales Data Analysis and Forecasting System** that integrates data analytics, machine learning, and business intelligence techniques.

The proposed system analyzes Walmart retail sales data to identify trends in sales performance, customer behavior, and product demand. The system applies **Exploratory Data Analysis (EDA)** techniques to extract insights and uses **machine learning models such as ARIMA and Linear Regression** for sales forecasting. The processed data is visualized using **Power BI dashboards** to provide interactive insights for business decision-making.

Additionally, a **Streamlit web application** is developed to deploy the analytics system and provide a user-friendly interface for accessing insights and predictions. The system helps identify peak sales hours, top-performing branches, and customer preferences. Forecasting results help businesses plan inventory, optimize staffing, and improve marketing strategies.

The proposed solution provides an end-to-end sales analytics system that supports **data-driven decision-making** and improves operational efficiency. The results demonstrate that combining machine learning and business intelligence tools enhances retail sales analysis and forecasting accuracy.

KEYWORDS

Retail Analytics, Machine Learning, Sales Forecasting, Power BI, Data Science, Streamlit

STATEMENT OF ORIGINALITY

This research paper presents an end-to-end Walmart Sales Data Analysis and Forecasting system using machine learning and business intelligence tools. The proposed system integrates data preprocessing, exploratory data analysis, forecasting models, and interactive dashboards. A Streamlit-based web application is developed to deploy the analytics system and provide user-friendly access to insights and predictions.

1. INTRODUCTION

Retail businesses generate a large volume of transactional data from daily sales activities. Analyzing this data helps organizations understand sales patterns, customer preferences, and product demand. However, traditional reporting systems often fail to provide meaningful insights, making it difficult for businesses to make data-driven decisions.

With advancements in data science and machine learning, businesses can analyze historical data to identify patterns and predict future trends. Retail companies such as Walmart operate across multiple branches and handle thousands of transactions daily. Therefore, automated analytics systems are required to improve operational efficiency.

This research presents a **Walmart Sales Data Analysis and Forecasting System** that integrates machine learning and business intelligence techniques. The system analyzes sales data, identifies trends, and predicts future sales.

2. LITERATURE REVIEW

Retail sales forecasting and analytics have gained significant attention in recent years due to the increasing availability of large-scale transactional data. Several studies have focused on applying data analytics and machine learning techniques to improve retail performance and forecasting accuracy.

Smith et al. (2020) proposed a time-series forecasting model using **ARIMA** for retail sales prediction. The study demonstrated that ARIMA models effectively capture seasonal trends and provide accurate predictions for short-term forecasting. The research highlighted the importance of time-series analysis in retail decision-making.

Brown et al. (2021) applied **Linear Regression and Machine Learning algorithms** to predict retail demand. Their research showed that regression-based models are useful for identifying relationships between sales and influencing factors such as customer demand and product pricing.

Lee et al. (2022) developed a **Business Intelligence Dashboard** to analyze retail sales data. Their study demonstrated that interactive dashboards improve decision-making by providing visual insights into sales trends, customer behavior, and product performance.

Kumar et al. (2023) proposed a **Machine Learning-based predictive analytics system** for retail businesses. The study used forecasting models to improve inventory planning and demand prediction. The results showed improved operational efficiency and reduced stock management issues.

Several researchers have also explored the use of **data visualization tools such as Power BI** for business analytics. These tools enable interactive data exploration and help stakeholders make informed decisions.

Despite these advancements, many existing systems lack integration between **data analytics, forecasting models, and dashboard visualization**. This research addresses this gap by developing a **complete end-to-end analytics system** that combines data preprocessing, exploratory data analysis, forecasting models, and interactive dashboards.

The proposed system improves retail sales forecasting and provides actionable insights for business decision-making.

3. METHODOLOGY

This research proposes an **end-to-end Walmart Sales Data Analysis and Forecasting System** that integrates data analytics, machine learning, and business intelligence techniques. The methodology followed in this project consists of multiple stages, including data collection, preprocessing, exploratory data analysis, forecasting, and visualization.

3.1 Data Collection

The dataset used in this project is the **Walmart Sales Dataset**, which contains transactional data from retail operations. The dataset includes various attributes such as:

- Invoice ID
- Branch
- City
- Customer Type
- Gender
- Product Line
- Unit Price
- Quantity
- Total Sales
- Date and Time
- Payment Method
- Rating

This dataset is used to analyze sales trends and develop forecasting models.

3.2 Data Preprocessing

Data preprocessing is an important step in data analysis. The dataset was cleaned and prepared for analysis using Python libraries such as Pandas and NumPy.
The following preprocessing steps were performed:

- Handling missing values
- Removing duplicate records
- Converting date and time formats
- Feature engineering (Month, Day, Hour)
- Data validation and consistency checking

These steps improved data quality and prepared the dataset for further analysis.

3.3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand sales trends and patterns. The analysis focused on:

- Branch performance analysis
- Product performance analysis
- Customer behavior analysis
- Payment method analysis
- Time-based sales trends

Various charts and visualizations were created to extract meaningful insights from the dataset.

3.4 Machine Learning Models

Machine learning models were implemented to predict future sales trends. The following models were used:

ARIMA Model

The ARIMA model is used for time-series forecasting. It analyzes historical sales data and predicts

future sales.

Linear Regression

Linear Regression is used as a baseline forecasting model. It identifies relationships between time and sales.

These models were trained using historical sales data and evaluated using performance metrics.

3.5 Dashboard Development

Power BI was used to develop interactive dashboards. The dashboard provides insights into:

- Sales performance
- Branch comparison
- Product demand
- Customer behavior
- Payment methods

The interactive dashboard allows users to explore data dynamically.

3.6 Web Application Deployment

A Streamlit web application was developed to deploy the analytics system. The web application provides:

- Interactive dashboard
- Forecasting results
- Business insights

The methodology ensures a complete **end-to-end analytics pipeline** from data collection to deployment.

4. SYSTEM ARCHITECTURE

The **Walmart Sales Data Analysis and Forecasting System** follows a structured architecture that integrates data processing, machine learning, and visualization components. The system is designed to convert raw sales data into meaningful insights and predictions.

The architecture consists of the following modules:

4.1 Data Collection Module

The first stage involves collecting the Walmart sales dataset. The dataset contains transactional data including product details, customer information, and sales values. This dataset forms the foundation of the analysis process.

4.2 Data Preprocessing Module

The collected dataset is cleaned and prepared for analysis. The preprocessing module performs the following operations:

- Handling missing values
- Removing duplicate records
- Feature engineering
- Date and time conversion

This module ensures that the dataset is ready for analysis and forecasting.

4.3 Exploratory Data Analysis Module

Exploratory data analysis is performed to identify patterns and trends. The analysis includes:

- Branch performance
- Product performance

- Customer behavior
- Payment analysis
- Time-based trends

These insights help understand business performance.

4.4 Machine Learning Module

Machine learning models are used for forecasting future sales trends. The models used include:

- ARIMA Model
- Linear Regression Model

These models analyze historical sales data and predict future sales.

4.5 Visualization Module

Power BI dashboards are created to visualize insights. The dashboard includes:

- Sales overview
- Branch performance
- Product performance
- Customer analysis
- Payment analysis

4.6 Deployment Module

The final module involves deploying the system using Streamlit. The Streamlit application provides an interactive interface for users.

Architecture Flow

Walmart Dataset
↓
Data Preprocessing
↓
Exploratory Data Analysis
↓
Machine Learning Models
↓
Power BI Dashboard
↓
Streamlit Web Application

This architecture ensures scalability and modular design.

5. IMPLEMENTATION

The **Walmart Sales Data Analysis and Forecasting System** was implemented using Python, Power BI, and Streamlit. The implementation process involved multiple stages.

5.1 Data Preprocessing Implementation

Python libraries such as Pandas and NumPy were used for preprocessing. The following steps were implemented:

- Data cleaning
- Feature engineering
- Data transformation
- Missing value handling

5.2 Exploratory Data Analysis Implementation

Exploratory data analysis was performed using visualization libraries. The following analyses were performed:

- Sales trend analysis
- Branch performance analysis
- Product performance analysis
- Customer behavior analysis
- Payment analysis

Charts were created using Matplotlib and Seaborn.

5.3 Forecasting Model Implementation

Machine learning models were implemented using Python:

ARIMA Model

Used for time-series forecasting.

Linear Regression Model

Used as baseline forecasting model.

Model performance was evaluated using:

- MAE
- RMSE

5.4 Dashboard Implementation

Power BI was used to create interactive dashboards including:

- KPI Dashboard
- Sales Trend Dashboard
- Product Analysis Dashboard
- Customer Analysis Dashboard

5.5 Deployment Implementation

Streamlit was used to deploy the system as a web application. The web app includes:

- Interactive dashboard
- Forecast results
- Data insights

The implementation successfully integrates data analytics, machine learning, and visualization

6. RESULTS AND DISCUSSION

The **Walmart Sales Data Analysis and Forecasting System** successfully analyzed the retail dataset and generated meaningful business insights. The results obtained from exploratory data analysis and forecasting models are discussed below.

6.1 Sales Performance Analysis

The total revenue generated from the dataset was approximately **₹155.08K**, with **3K units sold** and **499 customers**. The analysis showed steady growth in sales performance, indicating stable business operations.

The gross margin was approximately **5%**, which suggests that although revenue generation is stable, there is room for improving profitability.

6.2 Branch Performance Analysis

The branch-level analysis revealed:

- Branch C recorded the highest revenue
- Branch A and Branch B showed similar performance
- Mandalay city contributed the highest sales

These insights indicate that Branch C performed better than other branches and can serve as a benchmark for improving other locations.

6.3 Product Performance Analysis

Product line analysis showed:

- Food and Beverages generated highest revenue
- Sports and Travel performed strongly
- Electronic Accessories showed consistent demand

The balanced distribution across product lines indicates stable demand across categories.

6.4 Customer Behavior Analysis

Customer analysis showed:

- Female customers contributed slightly higher purchases
- Member customers generated higher gross income
- Average customer rating was around 6.98

These results indicate moderate customer satisfaction and opportunities for improving customer experience.

6.5 Payment Method Analysis

Payment analysis revealed:

- Cash payments accounted for significant portion
- E-wallet usage increased
- Credit card payments remained stable

This indicates growing adoption of digital payment methods.

6.6 Forecasting Results

The forecasting models predicted future sales trends based on historical data. The ARIMA model provided reliable short-term predictions, while Linear Regression provided baseline forecasting. These predictions help businesses plan inventory and improve decision-making.

7. CONCLUSION

This research presented a **Walmart Sales Data Analysis and Forecasting System** that integrates data analytics, machine learning, and business intelligence tools. The system successfully analyzed retail sales data and generated meaningful insights.
The project identified key insights such as:

- Top-performing branches
- Popular product categories
- Customer purchasing behavior
- Payment method trends
- Peak sales hours

Machine learning forecasting models predicted future sales trends, helping businesses improve planning and decision-making.
The Power BI dashboard provided interactive visualizations, and the Streamlit web application improved accessibility. The system demonstrated how data science can be applied to solve real-world retail problems.
Overall, the proposed system provides an **end-to-end retail analytics solution** that supports data-driven decision-making.

8. FUTURE WORK

The system can be improved with additional features in the future:

8.1 Real-time Data Integration

Real-time sales data can be integrated for live analytics.

8.2 Advanced Machine Learning Models

Advanced models such as LSTM and Prophet can improve forecasting accuracy.

8.3 Cloud Deployment

The system can be deployed on cloud platforms like AWS and Azure.

8.4 Customer Segmentation

Customer segmentation models can improve marketing strategies.

8.5 Inventory Optimization

Forecasting models can be used for stock optimization.
These improvements will enhance system capabilities.

9. REFERENCES

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